

		$\sqrt{\langle c^2 \rangle}$ at T1 $\sqrt{PV}$ at T1 $\sqrt{1}$
13	D	Once gas is released, it expands and achieves atmospheric pressure eventually. Using ideal gas equation, pressure is same before and after, but new temperature is 80°C $\frac{V}{T} = \frac{V'}{T'}$ $\frac{V}{273+27} = \frac{V'}{273+80} \rightarrow V' = \frac{353}{300} V$ Since gas expands to $\frac{353}{300}$ of original volume, the volume of gas remaining in the original bottle is still V, which means $\frac{V}{\frac{353}{300}V} = \frac{300}{353}$ of the original gas remains in the bottle which is 85%
14	B	Total Energy = $\frac{1}{2} m x_0^2 \omega^2 = \frac{1}{2} (0.60)(0.15)^2 (2\pi/5.0)^2 = 1.1 \times 10^{-2} \text{ J}$
15	A	Further damping causes amplitude to decrease at all frequencies